2.1 FULLY AUTOMATIC MIXER (3000 LITRES CAPACITY) AND STORAGE TANK (3000 LITRES CAPACITY) WITH ACCESSORIES INCLUDING ELECTRIC PANEL

2.1.1. Mixing Vessel – 3000 LTR

GENERAL	
Function	To prepare Liquid.
Design	BuiltasperGoodEngineeringPracticeandtocomplycGMP standard.
Material Of Construction	SS 316 contact parts and SS 304 non contact parts.
Type	Cylindrical Vertical with welded top / bottom dish end
Mounting	Vertical on Legs with option for load cell
PROCESS DETAILS& OPERATIONAL DATA	
Operation Type	Batch Type
Specific Gravity	0.92gms/cc
Operating Temperature	Ambient
Operating Pressure	ATM + Static
Minimum Steerable Volume	20 % of Gross Volume
DESIGN DATA	
Working Volume	3000 Ltr.
Gross Volume.	3600 Ltr
Type of Dish end	10% Torri spherical top / bottom dish ends.
Inner Shell ID / Shell Ht.	ID Dia. 1460 x 1875 mm (WL/WL)
Thickness of Shell / Dish end, mm	5.0 / 6.0 mm
Joint efficiency	0.85
No. of Legs	04 Nos. Ss Pipe Legs.
Design temperature	100 Deg. C
Design Pressure – Vessel	ATM + Static
Body Flange	Not applicable, as top dish is welded with Shell.
MATERIAL OF CONSTRUCTION	
Shell / Bottom / Top dish	Stainless Steel 316 Grade
Stirrer Mounting Pad	StainlessSteel304GradewithStainlesssteel316grade bushing.
Stirrer Shaft	Stainless Steel 316 Grade
Gasket	Food grade Silicon
Nozzles	Stainless Steel 316 Grade, Seamless pipe
Mechanical Seal Type	Single Dry Mechanical seal
Baffles	Stainless Steel 316 Grade
Ladder	Required
Legs Pipe	Stainless Steel 304 Grade

Base plate	Stainless Steel 304 Grade
Hardware – Internal	Stainless Steel 316 Grade
Hardware – external	Stainless Steel 304 Grade
Product outlet Valve type / MOC	SanitaryDesignwithTCconnection/StainlessSteel316 Grade
DRIVE AND AGITATOR DATA	
Type of agitator	Baby Anchor with 2 Blade PBT
Agitator Entry	Top Entry
Mixing speed	60 minutes for 3000 litres
No. of Agitator	As per mixing speed
Agitator Design	As per design
Shaft Dia.	80 Dia. In two pieces
Shaft Type	Solid in two piece
RPM	As per mixing speed
No. of Fix Baffles	As per design
Motor Type	TEFC/Non FLP / Flange Mounted / 3Ph/415V/AC/ IE 2 or equivalent
Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent
Transmission	RigidCouplingforshaftjointinsidethevesselandFlexible couplingforgearbox–agitatorshaft.
	Motor to Gear Box – direct couple
Shaft Sealing	Single Dry Mechanical seal
Gear Box for Main Drive	In Line Helical / Worm reduction type
Bearing Housing	CI/MS Bearing housing with heavy duty Taper roller bearings – 02nos. or equivalent
Drive Stool assembly	Mild Steel stool with SS 304 cover
Agitator PAD Plate	Solid SS 316 with SS 316 bushing.
Scrapper Blades	PTFE scrapper blades shall be provided to cover total wall surfaces of the mixingvessel.
NOZZLES - TOP DISH	
Man Hole	450 mm with dished cover and 100 mm sight glass
Light Glass	100 MM Pad Type with FLP Vessel lamp
Inlet	40 MM with valve
Breather Valve with Flame arrestor	40 mm with breather valve
Spare	25 mm with Valve – 02 nos.
CIP	Provided with spray ball
SHELL	
Temperature Sensor	½” BSP
Phase Inlet	40 mm with valve

Recirculation	50 mm with Valve
BOTTOM DISH	
Product Outlet	50 MM with ball valve
Filter	Suitable filter (<200 micron)
SURFACE FINISH	
Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra
Product non contact parts	Polished to matt finish. Surface roughness < 0.8 Ra
Mild Steel Surfaces	Two Coat of Zinc phosphate epoxy primer
	Final Coat of light gray acrylic poly urethane enamel paint

- Fixation of the mixer plant at 15 feet and above height of the building.
- Suitable electrical panel board with voltage switch control like MCB.
- Required pipelinesin SS316, electrical supply to transfer the liquid to storage tank.
- Graduation of the tank should be indicated with water bulb in outside of the tank for loading water and raw materials.
- Electronic weighing system for the weigh hopper to feed raw materials into the mixer tank.
- **Required – One no.**

2.1.2. Storage Vessel – 3000 LTR

GENERAL	
Function	Storage of Liquid.
Code	Built as per Good Engineering Practice and to comply cGMP
MOC	SS 316 contact parts and SS 304 non-contact parts.
Vessel Type	Cylindrical Vertical with welded top / bottom dish end
Process and Operation Data	
Fluid Handled in Vessel	Liquid
Service	Storage
Operating Temp.	Ambient
Operating Pressure of Vessel	ATM + Static
Operating Type	Batch
Gross Volume (L)	3300 Ltrs
Working Volume (L)	3000Ltrs
Nozzles	Dia450mm, ManHole 50 mmInlet, 40 mm Spare25 mm outlet with valve
DESIGN DATA	
Type of Dish end	5% Torri spherical top / bottom dish ends.

Joint Efficiency	0.7
Main Shell ID	ID Dia. 1575 mm
Height of Shell / Cone	1500 / 500
Thickness of main shell	4 mm
Thickness of top lid	4 mm
Base Plate	150 x 150
Legs Pipe	Dia.100 NB x 4 Nos.
AGITATOR DATA (SUBJECT TO REVALIDATION OF DESIGN DATA)	
Type of agitator	Baby Anchor with 2 Blade PBT
Agitator Entry	Top Entry
Mixing speed	Slow mixing to avoid sedimentation
No. of Agitator	As per mixing speed
Agitator Design	As per design
Shaft Dia.	As per mixing speed
Shaft Type	As per mixing speed
RPM	As per mixing speed
No. of Fix Baffles	As per design
Motor Type	TEFC/Non FLP / Flange Mounted / 3Ph/415V/AC/ IE 2 or equivalent
Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent
Transmission, shaft sealing and all other requirements	To be met out for slow mixing to avoid clumps formation.
MATERIAL OF CONSTRUCTION	
Product Contact Parts	Stainless Steel 316 Grade
Product non contact parts	Stainless Steel 304 Grade
Outlet Ball Valve	Stainless Steel 316 Grade – 25 with TC end
Bolts & Nuts – Internal	Stainless Steel 316 Grade
Bolts & Nuts – External	Stainless Steel 304 Grade
BOTTOM DISH	
Product Outlet	50 MM with ball valve
Filter	Suitable filter (<200 micron)
SURFACE FINISH	
Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra
Product non contact parts	Polished to matt finish. Surface roughness < 0.8 Ra
Mild Steel Surfaces	Two Coat of Zinc phosphate epoxy primer
	FinalCoatoflightgrayacrylicpolyurethane enamelpaint

- Fixation of the storage plant at ground level of the building.
- Suitable electrical panel board with voltage switch control like MCB.

- Required pipelines in SS316 and suitable motor to transfer the liquid to the bottle filling machine storage tank.
- Graduation of the tank should be indicated with water bulb in outside of the tank for loading water and raw materials.
- **Required – One no.**

2.2. FULLY AUTOMATIC LIQUID FILLING IN BOTTLE AND INNER PLUGGING AND OUTER CAPPING (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART

2.2.1. Turn table/Bottle feeder/Container Loader

Turn Table should be supplied with Standard Variable Speed Pulley Drive or A.C Variable Frequency Drive System for smoother Start and Stop operation for unstable products. The unit should be built on S. S. Square pipe and totally encompassed with S. S. Coverings. The S. S. cladded aluminium cast disc, soiled guide rails; helical gear housing provides a serials and noiseless performance for a long time.Turn Table should be ideal for use in automatic high-speed liquid line and turn table should also work as a buffer between two machines. If any of machines stop during the online operation, Turn Table can collect the outfeed product from machine or similarly can feed the products to machine without interrupting the complete line.

Operation Requirements

- Container inputs in turn table by manually and automatically will rotate on disk of turn table and exit through outlet, during rotation of Container will guided by a SS strip, which will guide the container towards outlet path.
- The variable pitch pulley ensures total and finer synchronization for incoming and out-going Glass Containers.

Technical Specifications

Machine type	Automatic turn
Direction	Clockwise / Anti clock wise as per requirement
RPM of Disk	4/5/6 through step pulley or equivalent to meet the required speed
Power and Motor	0.5 to 1.0 HP/220 to 415 V, 50 to 100 cycles/ second or equivalent

Output	100 to 120 Product /Minute (Required speed in filling line, 60BPM)
Working Height	800 mm to 910 mm according to the requirement

Salient Features

- All exposed parts are of S.S. 304 to prevent rusting.
- Reduction in gear ensure jerk free & noise less performance
- Allows absolute cleanliness even inside.
- Square pipe frame imparts vibration free performance and allows absolute cleanliness and ease of cleaning even from inside of the unit.
- Required conveyor from auto turn table to air-jet cleaning machine to maintain the speed.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- **Required – One no.**

2.2.2. Automatic Vertical Air-jet Cleaning Machine

Air jet cleaning machine should be made compact, versatile and enclosed in stainless steel elegantly matt finish body. This unit should consist of S.S. Slat Conveyor & Reciprocating cleaning nozzles. There should not be cleaning system arrangement when there is no container. The main drive of the machine and conveyor drive consist of A/c Motor with synchronized variable A/c Frequency Drive.

Technical Specifications

Machine type	Automatic
Out put	60 Bottles per Minute
No. of Cleaning Nozzles	According to the 60 bottles out put
Direction of Movement	Left to Right
Motor Specification	0.5 HP / 415 Volts / 50 Hz. 0.5 HP x 2 = 2 HP / 415 Volts / 50 Hz. or equivalent
Cleaning Cycle	Positive air pressure and vacuum cleaning
Working Height	800 mm to 910 mm according to the requirement

Salient Features

- Easy operation
- Stationary SS nozzles
- Automatic loading and unloading of container
- Individual control for air pressure and vacuum cleaning
- Higher capacity blower for proper cleaning
- SS Pressure met
- Required conveyor from auto turn table to filling line to maintain the speed.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- **Required – One no.**

2.2.3. Automatic volumetric liquid filling machine in bottles

The automatic volumetric liquid filling machine should be made compact, versatile and enclosed in stainless steel elegantly matt finish body. All parts should be made from **SS-316** Materials.This unit should work on volumetric principal and reciprocating self-centering device.

liquid Bottle size for reference

Bottle size – Height 205mm, Diameter – 235mm, Neck size – 80mm.

Filling volume – 600ml

Operation Requirements

- Nozzle should go upwards slowly from the bottom level of bottle towards neck during filling to minimize foaming adjustable nozzle is reciprocating according to filling dose.
- Machine with optional turn tables at in-feed and out-feed with individual drives to match the speed of incoming and outgoing bottles for smoother operation.
- The dosing block with a hexagonal bolt, this means different fill size can be set easily within minimum of time.
- The main drive should consist of a helical gearbox driven by A/c motor & through controlled by an AC frequency drive. The speed can be set in terms of bottles per minute. The conveyor drive shouldconsist of a hallow shaft, geared motor controlled by an AC frequency drive. A knob should be set for controlling the speed of conveyor.

Technical Specifications

Machine		Automatic
Output		Up to 60 Bottles per minute
Filling Volume		600ml
Direction of Movement		Left to Right
Number of head/syringe		According to the output of 60 BPM
Electrical Specification	Main Machine	2.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM
	Conveyor	1.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM
Change parts required		(A) Syringes. (B) Nozzle (C) Inlet & outlet hose pipe
Storage tank		SS316 with the capacity of more than 500 litres
Working Height		800 mm to 910 mm according to the requirement

Salient Features

- Suitable for bottle size700ml.
- Output: 60-65 Bottles/Minute.
- Filling volume – 600ml.
- Special type diving nozzle system to prevent against foaming problems.
- Nozzle lock system to stop dropping problems.
- Total contact parts should be SS – 316.
- Automatic speed control system.
- Automatic sensor-based bottle counter system.
- Bottle stopper and holder pneumatic based.
- Rigid vibration free construction for trouble free performance.
- Elegantly matt finished body.
- Minimum change overtime, from one size to another fill size.
- Self-lubricating for SS slat chain provides low friction wear surface, smooth and noiseless conveying.
- Safety guard for conveyor gear box, motor and pneumatic cylinders inbuilt protection against over current, Voltage fluctuation, short circuit in
- SS conveyor belt made from the SS316 (Lengthas per the speed of bottles filling – 60BPM) Adjustable Conveyor height as per choice to align with another line machine.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.

- **Required – One no.**

2.2.4. Automatic outer plugging machine

The automatic outer plugging machine should be made compact, versatile and enclosed in stainless steel elegantly matt finish body. Suitable to capping inner caps output - 60 Bottles/Minute (Bottle outer cap size: Height – above 25mm, Diameter – Above 100mm). One complete set of change parts suitable for any one size diameter of caps & bottle. Total machine construction in SS-304 wherever; Contact parts - SS-316 material.

Operation Requirements

- Outer capshould be placed according to the 60 bottles filling speed.
- Machine with optional turn tables at in-feed and out-feed with individual drives to match the speed of incoming and outgoing bottles for smoother operation.
- The speed can be set in terms of bottles per minute.

Technical Specifications

Machine Type	Automatic
Type of machine	Rotary / Straight according to the output of 60BPM
Number of heads	According to the output of 60 BPM
Machine output	Up to 60 Bottles per minute
Direction of Movement	Left to Right
Powder characteristics	440 Volts, IIIPhaseor equivalent for output of 60BPM
Main machine drive, Motor and Gear box	Standard suitable model
Working Height	800 mm to 910 mm according to the requirement
MOC of machine frame	Mild Steel powder coated and outside cladding SS 304
MOC of Contact Parts	Stainless Steel 316
Machine Covers	Stainless Steel 304
MOC of all vital shafts	EN 31 / Stainless Steel 304/316
MOC of Conveyor	Conveyor Slat Stainless Steel 304 Slat with Nylon Strips on Side Conveyor Shafts Mild Steel with Hard Chrome or Zinc Plating Conveyor Sprockets Mild Steel or Nylon

Salient Features

- No Bottle − No Capping System.
- Machine construction - Stainless Steel Finish.
- Cap orientation automatic Start/stop system.
- In feed & Exit sat wheel with over load clutch system to avoid bottle breakage.
- Single motor synchronizes conveyor, star wheel & platform turret.
- Minimum change over time.
- Pick & place arrangement.
- Hopper should be completed dismantled for easy cleaning.
- Adjustable height conveyor belt.
- Low noise level & low power consumption.
- Suitable SS Conveyor for reaching outer cap screw holder.
- Sensor – To position inner cap and to stop when there is no cap.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- **Required – One no.**

2.2.5. Automatic Outer capping machine

Automatic Capping machine is versatile, self-supported on stainless steel leg with height adjustable adjustment system. The machine should have precision built on sturdy welded steel frame completely enclose in stainless steel sheet and doors are provided to facilitate the servicing of machine.Suitable to capping outer caps output - 60 Bottles/Minute (Bottle inner cap size: Height – 23mm and Diameter – 100mm). One complete set of change parts suitable for any one size diameter of caps &bottle. Total machine construction in SS-304 wherever; Contact parts - SS-316 material.

Operation Requirements

The feed container moving on conveyor belt should feed into an in feed star wheel through In-feed worm, star wheel bringing the container below the sealing head in the subsequent indexing part, meanwhile the bottle pickup a cap from the delivery chute of cap filling bowl, where the body and the neck of the container are positioned below the rotating head, where the sealing head should perform perfect operation of threading & sealing.

Technical Specification

Machine Type	Automatic
Output	Upto30-60BottlesperMinute
Option	With bowl or with cap elevator
Direction of Movement	Left to Right
No. of Sealing Head	According to the output of 60 BPM
Capping Unit	Magnetic Screw Capping Head with Torque Adjustment
Caps Feeding	Vibratory Bowl Feeder
Caps Hammering	Pneumatic System
Machine Motion	Rotary Motion -Index Based
No Cap in Shute Elevator Stop (If machine is with Elevator)	Sensor
Suitable for Products	PET / Glass/Aluminum/ HDPE Bottle
Application	To Tight & Seal the Caps on Bottles
Up-Down Movement	Mechanical (Cam System)
Vibrator Pillar Mounting	Main Platform
Speed Synchronization of Star Wheel & Capping Head	Variable Frequency Drive
Working Height	800 mm to 910 mm according to the requirement
Conveyor Length	Depending upon the product output of 60BPM
Change Parts	Required in case of change in Cap Size & Bottle Size

Machine's material of construction

Machine main frame	Square Pipe, Stainless Steel 304
Machine Coverings	Complete in Stainless Steel 304
Product Contact Parts	Stainless Steel 316
Non Contact Parts	Stainless Steel / Mild Steel / Aluminum / Teflon (Whichever is applicable)
Conveyor Slat	Stainless Steel 304 Slat with Nylon Strips on Side
Conveyor Shafts	Mild Steel with Hard Chrome or Zinc Plating
Conveyor Sprockets	Mild Steel or Nylon
Bearings	Suitable standard model
Plating (Wherever Applicable)	ENP / Zinc / Hard Chrome / Blackening
Capping Unit	Stainless Steel304
Bowl	Stainless Steel 304
Vibratory Bowl Feeder	Mild Steel Cladded in Stainless Steel 304
Vibrator Stand	Mild Steel with ENP Plating

Cap Elevator frame (If machine is withElevator)	Stainless Steel 304
Elevator Belt	PVC
Shute	Stainless Steel 304
Support Rod on Existing point of Indexing Star wheel	Stainless Steel 304
Star Wheel with Back Guide	12 mm thick UHMW / Nylon
Guide Brackets	Standard make – Plastic
Guide Size	100mm / 150mm
Guide Rod	Stainless Steel 304 (10mm Round Bar)
Slat Chain Pitch	3.5” Stainless Steel 304
Capping Head Pillar	Mild Steel with ENP / Hard Chrome

Machine’s electronics & pneumatics components

Capping Head Motor Make	1 H.P Standard Model
Capping Head Gear Box Make	Ratio: 20:1 Standard Model
Cap Elevator Motor Make (If machine is with Elevator)	Standard make
Cap Elevator Gearbox Make (If machine iswith Elevator)	Standard makeRatio: 20:1
Capping Head VFD Make	1 H.P Delta
Sensor Make	Standard make
Proxy	Pulse meter
“No Cap in Shute Elevator Stop” Sensor (If machine is with Elevator)	Standard make
Electrical Wiring	Ferrule Numbering on all the wires with ProperMarkings
Electrical Components	Standard make
Pneumatics	Standard make
Air Pipes	Standard make
Wires & Cables	Standard make
Control Panel	Relay Card (Push Button Type)

Machine’s power & air consumption requirement

Power	1 H.P / 230 V/AC or 1.5KW or equivalent
Phase	Single Phase/ Three Phase according to the requirement

Compressed Air Required	3 H.P Compressor (6 bar pressure 0.5 CFM) or equivalent
Vibrator	As per requirement

Salient Features

- Theunitismadecompact&versatile.
- S.S. Slatconveyor
- S.S. Elegantly Matt FinishedBody
- Built in front controlpanel
- VibratoryBowl
- With A/c FrequencyDrive
- No Bottle – No Capping System
- Low noise level & low power consumption
- Cap orientation auto. Start/stop system
- In feed & Exit sat wheel with over load clutch system to avoid bottle breakage.
- Single motor synchronizes conveyor, star wheel & platform turret.
- Minimum change over time.
- Pick & place arrangement.
- Hopper can be completed dismantled for easy cleaning.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- **Required – One no.**

2.3. FULLY AUTOMATIC LIQUID BOTTLES DRYING TUNNEL (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART

The automatic drying tunnel should be designed with precision, these machines are widely recognized for their durability, design excellence and high performance. It should be suitable for fast productivity and designed for high-speed production with zero maintenance and individual control system.

Technical Specifications

Machine Type	Automatic
Temp. Req.	100’ To 150’ Degree Centigrade in Chamber,
Heater	Heater With Heating System. Total 10 To 24 Kw
Motive Load	1.5 Hp Conveyor and 0.5 Hp Blower Fan Load
Drive	Variable A.C. Drive
Tunnel Open	According to speed
Panel	Separate Mounted Panel for Machine,
Insulation	Special High Density Cera wool Lining
Hear Radiation	For Out Put with Inner Reflectors
Speed	Up to 60 Bottles per minute
Direction of Movement	Left to Right
Power supply	240V /15A/50Hz/-1Ph or equivalent
Power consumption	03 Kw or equivalent

Material

- All Electrical components are siemens make or equivalent.
- All Switches are sulzer or equivalent.
- Flexible cable should be provided with PVC air hose protection.
- Separate amp meter for all three-phase circuit.
- Stud type connector for cable termination.
- Incoming MCB 32amp indo copper make or equivalent.
- Blower Motors are remi make or equivalent.
- Blower Motor are separately controlled.

Salient Features

- Programmable Temperature controller.
- Silicon coated Roller conveyor.
- High temperature wire mesh conveyor.
- Suitable for fast productivity, it is designed for high-speed production with zero maintenance.
- Individual control system for heating tunnel.
- Consistent and improved airflow in heating tunnel.
- Lower wattage air fine/Infrared heaters, Reduced power consumption.

- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- **Required – One no.**

2.4 FULLY AUTOMATIC LIQUID BOTTLES PACKING IN CARTON BOX UNIT WITH ACCESSORIES FOR ERECTION, PACKING, CLOSING AND STRAPPING FOR TRANSPORTATION – REQUIRED NO. 1 SET

Carton Box size for reference

Length - 470mm, Width – 390mm and Height – 270mm

2.4.1. S-Conveyor

Modular belt conveyor should be middle ground between belt conveyors and chain conveyors. A modular belt should consist of individual plastic modules usually linked together with joint rods. This modular belt should be driven by chain sprockets; the base frame of the conveyor should be constructed on standard profile supports. The modular belt should make it possible to accept hot workpieces or workpieces with sharp edges. Furthermore, large conveyor widths should be implemented without high pre-tensioning of the belts. However, in contrast to chain conveyors, this design should less maintenance-intensive and is a technically simpler solution. These systems are frequently used as cooling lines, since the chain modules can be designed to be permeable to air.

Technical Specifications

Machine Type	Automatic
Output per Hour	60 to 65 Bottle/Minute
Direction of Movement	Left to Right
Gear Box (Conveyor + Main)	Suitable Standard Make
Make Motor (Conveyor + Main)	Suitable Standard Make
Working Desk Height	860 To 910 mm according to the requirement
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.75 Kw according to the requirement
Speed	0-11 mtr/min

Salient Features

- It is one of the simplestconveyors.
- It is used for conveying crates, box, pouches, Bottles &bags.
- It can be moved 90deg, 180deg,270deg.
- It can carry various products up to100kg.
- Size of the conveyor is designed as per therequirement.
- Used in food & pharmaindustries.
- Easy to clean & maintain.
- Easy to replace belt as it ismodular.

2.4.2. Case Erector

Case Erectors are used to erect carton which are formed automatically and use tape as sealing materials to seal the bottom flaps of carton. A stack of corrugated blanks should be placed into the machine. The blanks are pulled out of the machine one-by-one and formed into cases. The newly formed cases are sealed at the bottom with tape or glue.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.75 Kw according to the requirement
Speed	0-10 mtr/min
Voltage	1P, 220V, 50Hzaccording to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm(Std machine)
Air supply	Air pipe φ10 mm,≥6Kgf/cm²
Noise	≤75DB
Material of construction	MS powder coated

2.4.3. Roller Conveyor

Roller conveyors are intended for transporting piece goods such as solid boxes or pallets with rigid, flat bases. The drive concept can be selected to suit our particular application. Whether driven by gravity, a tangential chain, a drive roller or a friction drive, all systems are available in straight lines and curves.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.37 Kw according to the requirement
Speed	0-20 mtr/min
Drive roller	SS-304
Structure	MS powder coated
Side guard	Aluminum guide
Foot	Suitable Standard Make
Pneumatic cylinder	Suitable Standard Make

Salient Features

- It is one of the drive roller conveyors.
- It is used in conveying crates & box in 90deg.
- It can carry various products up to 100kg.
- Size of conveyor is designed as per requirement.
- Used in different types of industries.
- Easy to clean & maintain.
- Easy to replace belt as it is modular.

2.4.4. Case Tapping Machine

Auto carton sealer/tapping is suitable for carton / case packing, which can be used for single use or production line. It adopts adhesive tape to seal carton/case. It can be finishing the sealing both top and bottom. Also, you can adjust the height and width for different cartons.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Phaccording to the requirement
Power consumption	0.75 Kwaccording to the requirement
Speed	0-20 mtr/min
Voltage	1P，220V，50Hz according to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm(Std machine)
Air supply	Air pipe φ10 mm,≥6Kgf/cm²
Noise	≤75DB

2.4.5. Pick and Place Unit

Typically mounted on a stable stand, pick and place robots are positioned to reach different areas to perform work. They use advanced vision systems to identify, grasp and move objects from one place to another. Matrix of bottle will pick by bottle gripper and place into the boxes as per no. of matrix pick and place can be done.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Phaccording to the requirement
Power consumption	0.37 Kwaccording to the requirement
Speed	0-20 mtr/min
Drag chain	Suitable Standard Make

Structure	MS powder coated
Linear guide	Suitable Standard Make
Foot	Suitable Standard Make
Pneumatic cylinder	Suitable Standard Make

2.4.6. Idler Roller Conveyor

After case sealing from top of the box filled box will come and accumulate on this gravity roller conveyor. From this point filled box will pick manually

Salient Features

- It is one of the gravity rollerconveyors.
- It is used for conveying crates,box
- It can be moved 90 deg, 180 deg, 270deg.
- It can carry various products up to 100kg.
- Size of conveyor is designed as perrequirement.
- For any industries we can useit.
- Easy to clean & maintain
- Easy is replace roller as it ismodular.

2.4.7. Carton box strapping machine

Auto carton box strapping is suitable for carton / case packing, which can be used for single use or production line. It adopts strap roll to seal carton/case with stoppage clips. It can be finishing the strapping at center and sides of the carton box according to the height and width of the carton box.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2-5/Minute or equal
Direction of Movement	Left to Right
Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement
Power consumption	400 watt
Strap width	Bopp tape5 -15 mm
Strap speed	1.8sec/strap M/c

Salient Features

- Box strapping machines are widely used to package all kinds and sizes of boxes including corrugated boxes and PP boxes.
- The semi automatic strapping machine with top belt driven, powered table is easy to maintain with easily openable table top. Strapping position adjustable by moving sensor position.
- Box Strapping Machines allow easy loading and unloading of strapping rolls and very carefully and efficiently strap various boxes.
- The number of boxes strapped, depends on the technical specification and size of the strapping machine.
- Some strapping equipment consists of electromagnetic clutch transmission for increased strapping speed and has an adjustable cooling time and strap tension and length for better performance.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- Required conveyor for manual feeding of carton boxes and also to send strapping machine.

2.5 SEMI-AUTOMATIC LIQUID BOTTLES PACKED CARTON BOX UNIT WITH ACCESSORIES FOR CLOSING AND STRAPPING FOR TRANSPORTATION

2.5.1. Top and Bottom drive type carton sealing machine

Auto carton sealer/tapping is suitable for carton / case packing, which can be used for single use or production line. It adopts adhesive tape to seal carton/case. It can be finishing the sealing both top and bottom. Also, you can adjust the height and width for different cartons.

Carton Box size for reference

Length - 470mm, Width – 390mm and Height – 270mm

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2 - 5/Minute or equal
Direction of Movement	Left to Right
Power supply	220Vpl Ac/50Hz according to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm (Std machine)

Salient Features

- Suitable for fast productivity, it is designed for high-speed production with zero maintenance.
- Individual control systemfor pasting the tape.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- Required conveyor for manual feeding of carton boxes and also to send strapping machine
- **Required – One no.**

2.5.2. Carton box strapping machine

Auto carton box strapping is suitable for carton / case packing, which can be used for single use or production line. It adopts strap roll to seal carton/case with stoppage clips. It can be finishing the strapping at center and sides of the carton box according to the height and width of the carton box.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2-5/Minute or equal
Direction of Movement	Left to Right
Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement
Power consumption	400 watt
Strap width	Bopp tape5 -15 mm
Strap speed	1.8sec/strap M/c

Salient Features

- Box strapping machines are widely used to package all kinds and sizes of boxes including corrugated boxes and PP boxes.
- The semi automatic strapping machine with top belt driven, powered table is easy to maintain with easily openable table top. Strapping position adjustable by moving sensor position.
- Box Strapping Machines allow easy loading and unloading of strapping rolls and very carefully and efficiently strap various boxes.
- The number of boxes strapped, depends on the technical specification and size of the strapping machine.

- Some strapping equipment consists of electromagnetic clutch transmission for increased strapping speed and has an adjustable cooling time and strap tension and length for better performance.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- Required conveyor for manual feeding of carton boxes and also to send strapping machine
- **Required – One no.**

2.6 AUTOMATIC BOTTLE PRINTER TO PRINT PLACE OF MANUFACTURE, DATE OF MANUFACTURE, BATCH CODE AND EXPIRY DATE

Automatic bottle printer should print the details of place of manufacture, date of manufacture, batch code and expiry date as per feeding into the bottle by sensor. It should be capable of printing in any text direction and any place of the bottle. It should be in permanent ink.

Technical specifications

Operating Temp	5 to 35 Degree C
Frequency	60 Hz
Humidity	10-80 % RH
Direction	Horizontal
Number of heads	1
Speed	3000-4000 label per hour
Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s
Voltage	90 V to 260 V,47 Hz to 63 Hz
Resolution	8 dots per mm
Print area	30mm to 50mm
Message Storage Capability	250
Maximum Lines of Print	5, 32pixel
Print Fonts	5, 7, 9, 12, 16, 19, 25, 32
Print Head Cleaning	Automatic

Printer ink	One set with machine and additional one set for further use

Electricals

- MCC Panel board With power cables and control cables
- Panel board, switches and cables – make and wire gauge should be mentioned.
- Electrification Material like Cables, Cable trays, Clamps, Earthing, G.I strips necessary for power wiring from Panel to respective Motors.
- Earthing, consisting of earth pits, running of earth wire etc for MCC panel and individual motor

Erection, Commissioning and trial run

- Complete erection of machines, drives, structure, Chequered plates and electrical till MCCpanel.
- Commissioning of total system toproduction.
- Minimum two months training to locally appointed staff for operating liquid production plant. Training should be free ofcost.
- Annual Maintenance charges should be free at least for first one year from the date of commissioning of feed plant.
- Any service work should be attended within 48hours.
- Guarantee and Warranty period should bespecified.
- Spare parts with price list should be submitted.

Structure

- Suitable structural design and quantities should be designed for standard production plant.
- Stability certificate should be provided after erection.
- Structural drawing for erection of machineries including platforms and ladder should be given.

Complete monitoring of plant by computerized Monitoring system

- PLC Panel.
- PLC control panel to control the above batching plant.
- Batch report generation and storage.
- Connectivity to PC for data transfer for printing.

- Desktop Computer.
- ***Rate should be quoted separately for complete monitoring system.***